Reciprocal Graphs

Exam code: 0607

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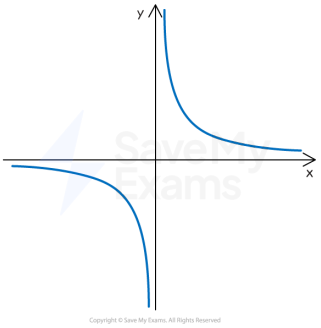
Reciprocal Graphs

What is a reciprocal graph?

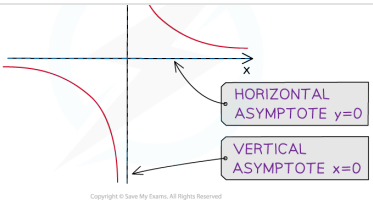
- A **reciprocal** graph is of the form $y = \frac{a}{x}$
  - It does **not** have a **y-intercept**
  - and does **not** have any **roots**
  - This means that the curve **does not cross** either the **x-** or **y-axes**



What are the asymptotes on a reciprocal graph?

- An **asymptote** is a line on a graph that a curve becomes closer and closer to but never touches
  - These may be **horizontal or vertical** lines
- A reciprocal graph has **two asymptotes**
  - A **horizontal** asymptote along the *x*-axis (with **equation** $y = 0$)
    - This is the **limiting value** of *y* when the value of *x* gets very large (either positive or negative)
  - A **vertical** asymptote along the *y*-axis (with **equation** $x = 0$)
    - This shows the **problem** of trying to **divide by zero**

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

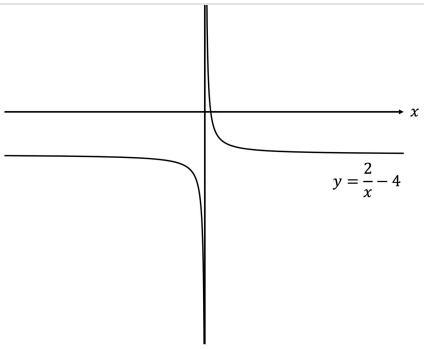


What about graphs of the form y = a/x + b ?

- The reciprocal graph, $y = \frac{a}{x} + b$ (where a and b are both constants)
  - is the **same shape** as $y = \frac{a}{x}$
  - but is **shifted upwards** by b units
    - $y = \frac{1}{x} + 2$ would be $y = \frac{1}{x}$ shifted **up** by 2 units
    - $y = \frac{4}{x} - 3$ would be $y = \frac{4}{x}$ shifted **down** by 3 units
  - This means the **horizontal asymptote** also **shifts up** by b units,
    - The **equation** of the horizontal asymptote is $y = b$
    - $y = \frac{5}{x} + 6$ would have a horizontal asymptote of $y = 6$
    - $y = \frac{1}{x} - 7$ would have a horizontal asymptote of $y = -7$
  - The **vertical asymptote** remains along the ***y*-axis**
    - The **equation** of the vertical asymptote is $x = 0$
    - $y = \frac{3}{x} + 9$ would have a vertical asymptote of $x = 0$

Worked Example

The diagram below shows the graph of $y = \frac{2}{x} - 4$.



(a) Write down the equation of the horizontal asymptote.

A reciprocal graph of the form $y = \frac{a}{x} + b$ will have a horizontal asymptote at $y = b$

$y = -4$

(b) Write down the equation of the vertical asymptote.

A reciprocal graph of the form $y = \frac{a}{x} + b$ will always have a vertical asymptote along the y-axis

$x = 0$

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